Robot motion control using Brain Computer Interface

R. Upadhyay, P.K. Kankar, P.K. Padhy, V.K. Gupta
Electronics and Communication Engineering
PDPM Indian Institute of Information Technology, Design and Manufacturing
, Jabalpur, India
(1110505@iiitdmj.ac.in,kankar@iiitdmj.ac.in,prabin16@iiitdmj.ac.in,vkgupta@iiitdmj.ac.in)

***Abstract*— In this work, two dimensional motions of a robot are controlled using brain computer interface. Motor Imagery signals for different mental activities are recorded using Electroencephalography technique. Recorded Electroencephalogram signals are filtered out for noise reduction and processed. Processed signals are further used to prepare the feature vector to train classifier algorithm. Appropriately trained and tested classifier algorithm is used to translate Electroencephalogram signals to a meaningful command. These commands are the electrical signals which further control the two dimensional motions of robot. For implementing Brain Computer Interface, Electroencephalogram signals are filtered out by Butterworth low pass filter and further preprocessed using Multi-scale Principal Component Analysis algorithm. Support Vector Machine classifier completes classification task.**

Keywords—Electroencephalogram, Butterworth filter, Multi-scale Principal Component Analysis, Support Vector Machine

I. INTRODUCTION

Brain Computer Interface (BCI) is an alternate pathway for communication between human brain and the outside world. BCI capabilities are largely helpful to the patient suffering from severe motor disorder to communicate and control assistive appliances and applications using brain power [1]. Major breakthrough in the field of BCI research was marked after invention of non-invasive techniques for EEG signals recording. Previously, electrodes and sensors were implanted inside the brain using invasive techniques. EEG signals recorded by using invasive techniques are less noisy due to less contamination from surroundings. But, user has to go under surgery for implantation. This makes invasive techniques an expensive and cumbersome approach. Various patterns of rhythmic or periodic activity exhibit by EEG signals can be divided in the frequency bands likewise 0.1- 3.5Hz (delta), 4-7.5Hz (theta), 8-13Hz (alpha), 13-30Hz (beta) and >30Hz (gamma) [2-3]. It is reasonable to filter out EEG signals above 30 Hz as most of the viable information is retained below 30 Hz of frequency. In this work, we have used EEG signals, recorded using non-invasive methods only. The essential steps for BCI realization are recording of brain electrical activity, preprocessing, feature extraction, classification and finally command generation. Successful working of a BCI is determined by careful measures taken during each step. Preprocessing the EEG signals combines filtration and noise removal from raw EEG data. Signals recorded using non-invasive techniques are more prone to noise. Common Spatial Pattern (CSP), Independent Component Analysis (ICA) and Principle Component Analysis (PCA) are very common and mostly used algorithms for EEG signals processing [4]. Neuromechanism employed for EEG signals recording affects largely the efficiency of BCI in terms of time and classification. Most commonly used neuromechanism are sensorimotor activity [5-7], Visual Evoked Potential (VEP) [8], Slow Cortical Potential (SCP) [9] and P300. EEG signals employed in present work are recorded for imagined motor activity using sensorimotor activity as neuromechanism for left fist blink, right blink and both leg movement.

In present work, EEG signals recorded for imagination of three different body part movements i.e. left fist blink, right fist blink and both leg movement are classified using trained SVM classifier for command generation to control two dimensional motion of a robot. For reducing complexity and processing time EEG signals recorded over C3 and C4 channels are only considered and further used. EEG signals are filtered using low pass Butterworth filter having cut off frequency 30 Hz and further processed using MSPCA model for reconstructing simplified EEG signals. MSPCA has been employed by many researchers for preprocessing and de-noising biomedical data and images like Zhang and Zhang [10] employed Gabor Wavelet based MSPCA and SVM for face recognition. A model of MSPCA by combining Discrete Wavelet Transform (DWT) and PCA for de-noising and decomposing EEG signals was proposed by Xie and Krishnan[11]. A Graphical User Interface is developed for training the classier and controlling the robot 2-D robot motion in off line mode to present the efficiency of model and working of hardware, which can be further replaced by a wheel chair for transportation of patient in real time.

II. AN OVERVIEW ON TECHNIQUES

A. Multiscale Principal Component Analysis

When behavior of the events associated with data changes rapidly with time and frequency, MSPCA is found be suitable modeling technique. Modeling of single scale data can be efficiently performed by PCA alone, as algorithm considers relationship of data for same time- frequency localization. However all practical data sources generate data which is multi-scale in nature, hence it is feasible to use PCA with multiscale capabilities. MSPCA performs computation of PCA on wavelet coefficients at each scale and combines results at significant scales [12]. In MSPCA algorithm, first the data is decomposed in time-frequency using certain wavelets. Then the PCA is applied for the coefficients of each scale irrelevant to one another. Combination of the models corresponding to important scales takes place to obtain a unified model [12].

B. Support Vector Machine

Vapnik developed SVM model in 1995 [13]. SVM model is a supervised machine learning technique. SVM is established on Statistical Learning theory. SVM model can be applied on both classification and regression problems for two or multiple

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classes of data. SVM model works on the principle of Empirical Risk Minimisation (ERM) and Structural Risk Minimisation (SRM). This makes SVM a strong classification and regression tool [14]. Due to many of these features offered by SVM, it has become first choice of researchers for classification and regression problems. A hyperplane classifies two different classes of data, by simply creating a boundary between them. For classifying multiple classes of data, multiple hyperplanes are created. Each hyperplane is orientated such a way that it maintains maximum distance from support vectors of two classes data, which ensures least generation error in the classification [15]. Support vectors are the closest data points to the hyperplane belonging to two classes of data.

Let sample data set be (xi, yi), with i= 1 to M training points. Each input xi has D attributes and belong one of two classes yi = -1 or +1. Solution of an optimization problem under some constraints, gives hyperplane f(x) = 0 to separate two classes of data. If w is normal to the hyperplane and $b/\|w\|$ is perpendicular distance from the hyperplane to the origin, expression for hyperplane can be written as:

$$y_i(x_i.w + b) - 1 \geq 0 \quad \forall_i \tag{1}$$

To maximize margin, following optimization problem is to be solved:

$$\text{Minimize } \frac{1}{2}\|w\|^2 \tag{2}$$

Such that

$$y_i(x_i.w + b) - 1 \geq 0 \quad \forall_i$$

In terms of Lagrange multiplier, the optimization problem can be written as:

$$\text{Maximize } W(\mu) = \sum_{i=1}^{M}\mu_i - \frac{1}{2}\sum_{i,j=1}^{M} y_i y_j \mu_i \mu_j x_i x_j \tag{3}$$

Such that

$$\begin{cases} \mu_i \geq 0 \\ \sum_{i=1}^{N} y_i \mu_i \end{cases} \qquad i = 1,2,\ldots,M$$

The overall problem is reduced to a convex quadratic optimization problem. The Sequential Minimal Optimization algorithm solve dual problem by decomposing quadratic programming problem into sub-problems [16].

III. BCI IMPLEMENTATION

A. Preprocessing

EEG data recovered from C3 and C4 channel has been filtered using low pass Butterworth filter, with cut-off frequency of 30 Hz, to eliminate noise associated with other frequency ranges. Essential information related to different brain activities concentrate below 30 Hz of frequency, so it is viable to filter out rest of the frequency components.

Filtered EEG signals are further preprocessed to simplify the EEG data matrix using MSPCA. For decomposition of EEG data, Symlets wavelet is selected and decomposition level was set to 5. Components associated with eigen values (Variance) greater than mean of all eigen values are retained, as per the Kaiser's rule. Fig.1 (a-b) shows the EEG data plots before applying and after application of MSPCA.

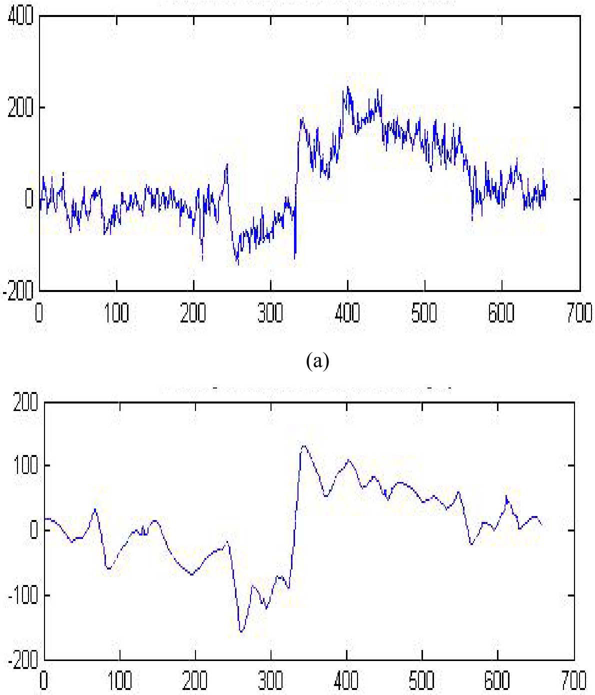

Fig. 1 (a) EEG signals for left filt moment imagery before applying MSPCA (b) Simplified EEG signals after applying MSPCA

EEG database is taken from PhysioNet ATM [17], an open source library for biomedical signals. EEG signals are measured from 64 channels and sampled at the rate of 256 Hz. The electrodes position is as per the international 10-20 system of electrodes placement [18].

On the basis of annotations information provided by PhysioNet for separate experimental runs, complete EEG data is divided into three categories or classes and used to prepare feature vector. EEG data used in this study can be categorized as follows:

1. Imagination of opening and closing of Left fist
2. Imagination of opening and closing of right fist
3. Imagination of both leg movements

Motor imagery is part of brain cortex, which is responsible for the planning, preparation and execution of any body movements, hence it is feasible to record EEG signals from C3 and C4 channels as channel position is above motor imagery.

B. Feature Extraction and Classification

Features are extracted from preprocessed EEG signals and feature vector is prepared to train classifier. Four Statistical parameters i.e. Skewness value, Kurtosis value, Standard Deviation and Mean value are calculated as features from processed EEG signals. In present work, SVM classifier is employed for classification or identification of the class of EEG signals. Once the trained classifier classifies provided input EEG signal, a command is generated according to the classifier output. Sample of feature input vector used for present work is present in Table. 1.

TABLE I. SAMPLE INPUT VECTOR FOR SVM

Features				Classes
Statistical features calculated processed EEG data				
Skewness	*Kurtosis*	*Standard Deviation*	*Mean*	
-5.466067	32.563071	1.9618785	-1.0069578	Left Fist blink Imagined
-5.072297	27.967893	2.0427659	-1.0481149	Left Fist blink Imagined
-5.399512	31.602019	1.9447994	-1.0050202	Left Fist blink Imagined
-5.347096	30.967038	1.5293819	-1.1418091	Right Fist blink Imagined
-5.365895	31.190113	1.5480403	-1.1452608	Right Fist blink Imagined
-5.407290	31.696262	1.5839810	-1.1507105	Right Fist blink Imagined
-0.702840	3.2173186	77.3435915	4.9375951	Both lag movement Imagined
0.278408	3.1685358	60.1114382	8.6544901	Both lag movement Imagined
-0.386435	2.8152150	165.897872	76.5038051	Both lag movement Imagined

C. *Working of hardware*

Offline motion controlling of a prototype robot is performed to show the complete working of proposed methodology. Methodology can be further tested and refined to work in real time for a wheelchair motion. A GUI is developed to demonstrate training, testing and command generation. GUI communicates with prototype robot and sends command over wireless channel for controlling motions. Two RF modules, one attached at serial communication port on transmitter side and another one with microcontroller on receiver side, establish a wireless communication channel between computer (GUI) and robot. Microcontroller receives commands from RF module and performs desired robot motion by taking necessary action for controlling two encoder motors attached to robot wheels. EEG signals for Left fist blinking turns robot in left direction. Likewise EEG signals for right fist blinking and both feet movements are used for turning robot right and straight directions correspondingly. Fig.2 shows the schematic of hardware module.

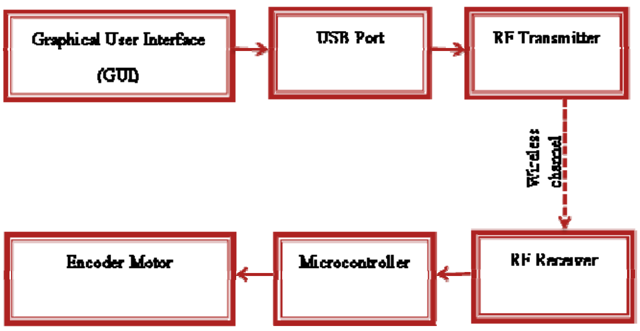

Fig. 2 Hardware module

D. *Graphical User Interface*

Description of GUI is as follows:

- Load and Train push button: Loads, preprocesses raw EEG data having three classes and trains SVM with extracted features for each class.
- Left signal push button: Load three EEG signals for testing, corresponding to class left turn.
- Right signal push button: Load three EEG signals for testing, corresponding to class right turn.
- Straight signal push button: Load features EEG signals for testing, corresponding to class straight movement.
- Pop up menu: Facilitates to choose one EEG signal out of three EEG signals of previously selected class for testing purpose.
- Plot: Shows the time domain plot of selected EEG signals.
- Generate command push button: Tests, selected EEG signal using SVM after extracting features. SVM identifies the EEG signal corresponding to one of the three classes. A command is generated and sent to the USB port for further transmission to the robot.
- Exit button: Closes GUI module.

E. *RF Module, Microcontroller and Motor Driver IC*

RF module includes RF Transmitter and RF receiver pair, which operate at the same frequency. Serial data is received by RF transmitter from USB, and RF transmitter transmits received serial data wirelessly over radio frequency. Transmitted data can be received by RF receiver operating at the same frequency. Fig. 3 and Fig. 4 shows RF module CC-2500 and GUI module developed for present work.

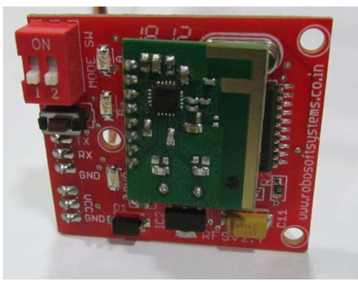

Fig. 3 RF module CC-2500

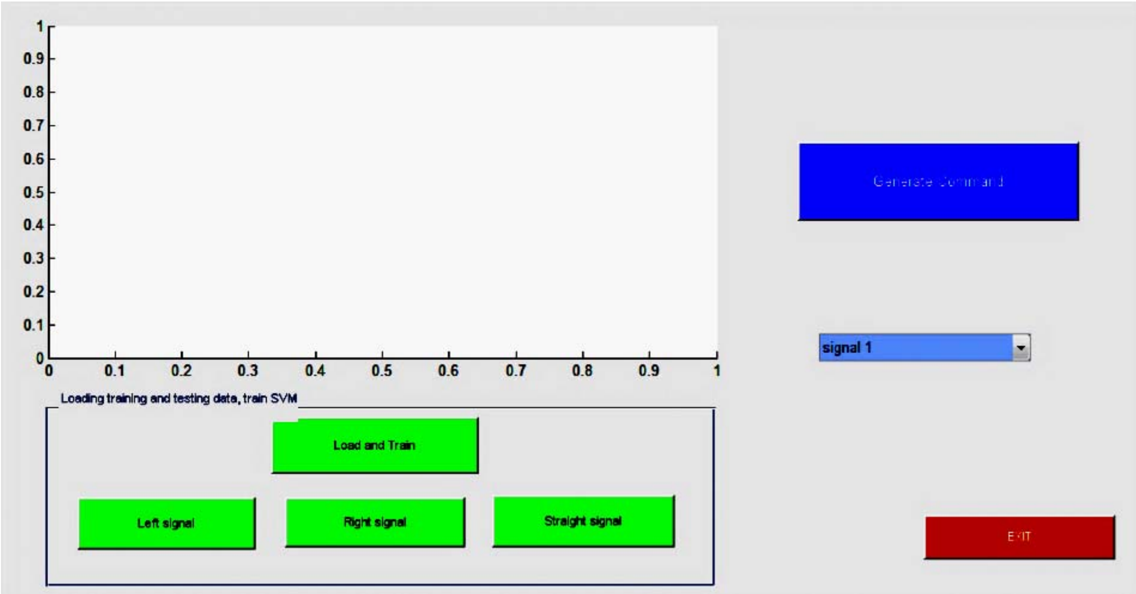

Fig. 4 GUI module

Atmega 16 microcontroller is used in this work to regulate encoder motors for achieving robot motion control. L298N Motor driver IC is used for driving two encoder motors. L298N has high voltage, high current ratings, this matches to the requirements of encoder motors. Fig. 5 shows the constructed robot.

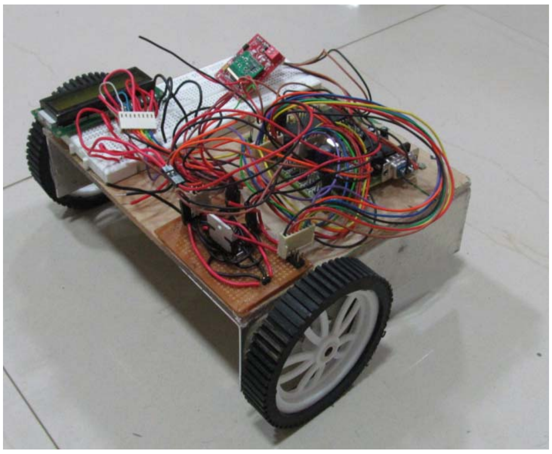

Fig. 5 Constructed robot

IV. RESULTS

EEG signals of a single subject from motor imagery, recorded for three different body part movements, have been used to train SVM classifier. For testing the efficiency of proposed methodology, 14% of the total EEG data unseen to training algorithm, is given to classifier. Table. 2 shows confusion matrix for classification of three classes of EEG data. From Table. 2, it is clear that, SVM can efficiently classify between mental imagination activity for left fist blink, right fist blink and both leg movement using input feature vector obtained from C3 and C4 channels. SVM classified three different movements with classification efficiency of 89% and generated respective command for robot motion. Wireless module successfully transferred the command for robot motion control.

TABLE II. CONFUSION MATRIX OF SVM CLASSIFICATION WITH MSPCA

Test Set			**Output**
left fist blink imagined	***right fist blink imagined***	***both leg movement imagined***	***Classified as***
6	0	3	left fist blink imagined
0	9	0	right fist blink imagined
0	0	9	both leg movement imagined

V. CONCLUSION

The aim of the study is to implement a BCI with three controlling commands for robot 2 dimensional motion controlling, using brain signal of imagination recorded only on C3 and C4 channel to reduce the complexity of multiple channels, and to input vector to SVM model consists of features which are extracted from pre-processed EEG signals for imagination of left fist blink, right fist blink and both lag movement. A methodology is proposed for the feature extraction from raw EEG data. For classification SVM classifier is used, which shows high classification efficiency for imagination task. BCI implemented successfully control 2-D robot motion in this work.

ACKNOWLEDGMENT

This work was supported in part by Department of Science and Technology, Government of India.

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